WNDRVR
SUPPORT NETWORK

VXWORKS 7 GETTING STARTED, SR0650

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VxWorks 7 Getting Started, SR0650

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1. OVERVIEW

The VxWorks® 7 Getting Started guide provides introductory information and links to topics related to installation, product documentation, development tools, and application development.

VxWorks 7 is a set of development tools and software run-time products that includes the VxWorks real-time operating system (RTOS).

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2. INSTALLATION

To help with the VxWorks 7 and WorkBench 4 installation process, Wind River provides an installation video, an administrator's guide, and a developer's guide to install and license Wind River products.

VxWorks 7 Host Requirements

For information on supported operating systems and hardware architectures for VxWorks 7, see the VxWorks 7 product release notes from the Wind River Support Network site at Getting Started.

VxWorks 7 Installation Video

To install VxWorks 7 based on the *Installing VxWorks 7* video instructions, you need the following:

- Access to an email from Wind River announcing the delivery of the product.
- Access to the Wind River Support Network site to view the video at Installing VxWorks 7.

The video instructions provide information on login operations to access the Wind River Support Network site and then WindShare where you can download, install, and update Wind River products including VxWorks 7 and Workbench 4.

Note: VxWorks 7 without Workbench 4 is not supported.

Wind River Product Licensing Video

For information on licensing Wind River products, go to Installation & Licencing on the Wind River Support Network site, select a License Administrator Tools version, and view the *Wind River Product Licensing* video.

Wind River Product Installation and Licensing Guides

For additional information on installing and licensing Wind River products, go to the Installation & Licencing and access the following documents:

- *Wind River Product Installation and Licensing Administrator's Guide*

  The administrator's guide outlines advanced configuration tasks for managing product licenses and installations for a workgroup.
- *Wind River Product Installation and Licensing Developer's Guide*

  The developer's guide outlines configuration tasks for managing product licenses and installations for a single workstation.

3. VXWORKS 7 DOCUMENTATION

The VxWorks 7 documentation is part of the Wind River Support Network and contains getting started guides, release notes, tutorials, programmer guides, API references, and user guides.

Wind River Support Network

The Wind River Support Network is the home to all technical and product information provided by Wind River.

The main page of the Wind River Support Network site provides an introductory video on how the site works and some of its key features. There is also a demonstration video on how to navigate and search within the Support Network.

All pages within the Support Network provide access to the knowledge forum, customer support, and enable you to provide feedback.

To access the Support Network from Workbench 4, navigate to **Help > Wind River Knowledge Library**.

VxWorks 7 Getting Started Documentation

The VxWorks 7 getting started documentation is available from the Support Network at Getting Started. The getting started documentation includes:

- VxWorks 7 Getting Started on page 1
- Video and text-based tutorials
- Release Notes
- Getting Started Defects
- Getting Started Downloads

VxWorks 7 Configuration and Build

For information on configuring and building VxWorks source build (VSB) projects and VxWorks image projects (VIPs), see the VxWorks 7 Configuration and Build Guide.

For information on how to collect recipe files from Workbench 4, clone an installation, and recreate the VSB and VIP project configurations, see the VxWorks 7 Configuration and Build Guide.

VxWorks 7 Layers and Package Management

VxWorks layers provide the infrastructure for the code that produces individual VxWorks features. Individual features, such as USB, file system, and BSPs, are built, tested, and released independently from each other using layers.

For more information on layers and packages, see the VxWorks 7 Layers and Package Management Guide.

Kernel Applications and Real-Time Process Applications

To access tutorials on creating kernel applications and real-time process (RTP) applications using Workbench 4 and the wrtool utility, go to Getting Started and select **Tutorials/Videos**.

For more information on creating, configuring, and building kernel application and RTP application projects, see the VxWorks 7 Kernel Application and RTP Application Projects Guide.

For more information on developing kernel applications and RTP applications, see the *VxWorks 7 Application Programmer's Guide:RTP Applications* and *VxWorks 7 Programmer's Guide: About Real-Time Processes*.

VxWorks 7 API Documentation

The following VxWorks 7 API references are available from the Support Network:

- Application Core OS
- Kernel Core OS
- Application Debug Analysis
- Kernel Debug Analysis
- Kernel Connectivity
- Kernel Multimedia
- Application Multimedia
- Kernel SNMP and Web CLI
- Application Open Source
- Kernel Open Source
- Application Networking
- Kernel Networking
- Application Storage
- Kernel Storage
- Application Security
- Kernel Security
- Kernel Virtualization Guest

Board Support Packages, Processors, and Device Drivers

VxWorks 7 provides a board support package (BSP) and driver model that allows the execution of VxWorks on custom hardware.

For information on supported BSPs, see VxWorks 7 Release Notes.

For more information on BSPs, processors, and device drivers, go to Processor and Device Support.

Networking

The VxWorks 7 Network Stack is a full-featured dual IPv4/IPv6 TCP/IP stack that supports multithreading, and simultaneous use of raw IP, UDP, and TCP over IPv4 and IPv6.

For more information on the network stack and network services, go to Networking.

Connectivity

VxWorks 7 supports the Controller Area Network (CAN) communications protocol, the IEEE 1394 serial bus, and the Universal Serial Bus (USB).

For more information on VxWorks 7 connectivity, go to Connectivity.

Graphics and Multimedia

VxWorks 7 supports audio driver framework, event device framework, and target-based graphics.

For more information on VxWorks 7 graphics and multimedia capabilities, go to Graphics and Multimedia.

Management Interfaces

VxWorks 7 supports CLI, Web, and MIBway to allow the creation of user interfaces for external or remote device management.

VxWorks 7 also supports the Simple Network Management Protocol (SNMP) including SNMP v1/v2c, SNMP v3, and SNMP AgentX.

For more information on management interfaces, go to Manageability.

VxWorks Simulator (VxSim)

The VxWorks simulator is a simulated hardware target for use as a prototyping and test-bed environment for VxWorks.

For more information on the VxWorks simulator, see the VxWorks 7 VxSim User's Guide.

App Designer Toolkit

The App Designer Toolkit is a set of tools to create custom-branded VxWorks Software Development Kits (VxSDKs). The App Designer Toolkit is an optional product for VxWorks 7.

For more information on the App Designer Toolkit, go to App Designer Toolkit.

VxWorks 7 Kernel

For more information on the VxWorks 7 kernel, see the *VxWorks 7 Application Programmer's Guide: VxWorks Header Files*.

Workbench 4 Documentation

The main page of the Workbench 4 documentation is at Workbench 4.

For video instructions on how to use Workbench 4 help, go to Workbench 4 and select **Getting Started > Tutorials/Videos > Workbench 4 Demo Series: Using Help**.

To get help for Workbench 4 from within the Workbench 4 integrated development environment (IDE), navigate to **Help > Help Contents** and select the **Wind River Documentation** topic.

To access the Wind River Support Network from the Workbench 4 IDE, navigate to **Help > Wind River Knowledge Library**.

4. VXWORKS DEVELOPMENT TOOLS

The VxWorks development tools include Workbench 4, the wrtool utility, and the VxWorks simulator (VxSim).

Workbench 4

Workbench 4 is an Eclipse-based development suite used to create real-time and embedded applications. It facilitates managing and building projects, establishing and managing host-target communication, and running, debugging, and monitoring VxWorks applications.

Workbench 4 contains perspectives, views, and editors to view, manipulate, and edit project components.

The Project Explorer view provides the ability to create, open, close, modify, and build projects. Various filters, sorting mechanisms, and viewing options help to make project management and navigation more efficient.

For more information on Workbench 4, go to Workbench 4.

wrtool

VxWorks 7 provides the wrtool utility to create, build, and manage VxWorks projects from the command-line interface (CLI). The wrtool utility starts Workbench 4 in a non-graphical mode, that is without the Workbench 4 graphical user interface.

The wrtool utility and Workbench 4 can be used interchangeably on the same project to create, build, and manage VxWorks projects.

For more information on the wrtool utility, see Project Creation with the wrtool Utility on page 12.

VxWorks simulator (VxSim)

The VxWorks simulator is a simulated hardware target for use as a prototyping and test-bed environment for VxWorks. The VxWorks simulator allows the development and test of VxWorks applications on a host system, reducing the need for target hardware during the initial phases of development. The VxWorks simulator supports the creation of a simulated target network for developing and testing complex networking applications.

For more information on the VxWorks 7 simulator, see the VxWorks 7 VxSim User's Guide.

Environment Variables

For information on setting environment variables for the VxWorks development tools, see Environment Variables for the VxWorks Development Tools on page 14.

5. VXWORKS PROJECT TYPES

Workbench 4 and wrtool support several different project types, but each one is used for a specific purpose. Choosing the appropriate project type for an application is the first development task.

VxWorks Source Build (VSB) project

Allows the configuration and build of VxWorks from source code.

The VxWorks 7 installation includes source code but does not include pre-built kernel libraries. The first step is to configure and build the VxWorks kernel libraries. The VxWorks kernel libraries enable the build of VxWorks image projects (VIPs).

VxWorks Image Project (VIP)

Allows the configuration and build of VxWorks from binary components. A VIP is based on a specific VSB project.

A VIP creates a VxWorks run-time kernel image based on components defined by component description files (CDFs). During the development cycle, you configure the VIP by selecting components and configuring parameters to support the application and development requirements.

After development, VxWorks can be reconfigured and rebuilt to support production systems.

Downloadable Kernel Module (DKM) Project

DKM projects enable the management and build of kernel application modules that execute in kernel space. Modules can be built separately from VIPs and debugged on a target running VxWorks, and loaded or unloaded as required.

Real-Time Process (RTP) Project

RTP projects allow the management and build of RTP application executables that run in user mode (in memory protected user space). Executables can be built separately from VIPs and debugged. At run-time, the executable file is loaded from the host or from a local VxWorks file system, and runs as an independent process.

RPM Project

RPM projects allow the creation packages in a multi-platform format for distributing complex, structured file sets, such as project source files or finished software products, to developers or end users. RPM-based installations ensure the integrity of common management tasks such as updates and package removal. The Package Manager provides the ability to add, remove, import, and build RPMs.

ROMFS File System Project

A ROMFS file system is a read-only memory file system that represents and stores files and directories in a linear way (similar to the tar utility). It is installed in RAM with the VxWorks system image at boot time. ROMFS provides the ability to bundle VxWorks applications, or any other files for that matter, with the operating system. No local disk or network connection to a remote disk is required for executables or other files. When VxWorks is configured with ROMFS, files of any type can be included in the operating system image simply by adding them to a **ROMFS** directory on the host system, and then rebuilding VxWorks. The build produces a single system image that includes both the VxWorks image and the files in the **ROMFS** directory.

Shared Library Project

Shared library projects are for libraries that are dynamically linked to real-time process (RTP) applications at run-time. You can store a shared library on a host file system, a network file system, or a local file system on the target (including a ROMFS file system).

Static Library Projects

Static library projects allow the creation of a library that can be statically linked to applications at build time. Use static kernel library projects to create a library that can be statically linked to kernel applications at build time. Use static user library projects to create a library that can be statically linked to real-time process (RTP) applications at build time.

Layer Project

VxWorks layers are containers for features including kernel code or additional user space libraries and processes that can be included into the VxWorks runtime. VxWorks layers typically consist of source code with configuration and build support. When configuring a VSB project, a layer is presented as a unique configurable feature.

For information on configuring and building VxWorks source build (VSB) projects and VxWorks image projects (VIPs), see the VxWorks 7 Configuration and Build Guide.

For information on creating, configuring, and building kernel application and RTP application projects, see the VxWorks 7 Kernel Application and RTP Application Projects Guide.

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6. VXWORKS DEVELOPMENT WORKFLOW

VxWorks development is highly variable, depending on the nature of the product. In all cases, you must build VxWorks from source code, and optionally configure VxWorks to suit the product.

The basic workflow is as follows:

1. Configure and build VxWorks.

   To configure and build VxWorks, proceed as follows:

   1. Configure and build the source code by creating a VxWorks source build (VSB) project.
   2. Configure and build VxWorks from binary components by creating a VxWorks image project (VIP) based on the VSB project.

   For information on how to configure and build VSB projects and VIPs, see the VxWorks 7 Configuration and Build Guide.
2. Create a downloadable kernel module (DKM) project for kernel-mode applications or a real-time process (RTP) project for user-mode applications.

   To access video and text-based tutorials on how to create, build, and run basic DKM and RTP applications using Workbench 4 and the wrtool utility, go to Getting Started and select **Tutorials/Videos**.

   For more information on how to create DKM and RTP projects and applications for VxWorks 7, see the VxWorks 7 Kernel Application and RTP Application Projects Guide.
3. Boot VxWorks and connect to a real or simulated target.

   For information on booting VxWorks, go to Processor and Device Support and select **Boot**.

   For information on how to connect to a target, go to Getting Started, select **Tutorials/Videos**, and view the *VxWorks 7 Demo Series: Using Workbench to connect to a VxWorks Target* video.

   For information on the VxWorks simulator, see theVxWorks 7 VxSim User's Guide .
4. Load and run an application.
5. Test and debug the application.
6. Package the project (optional step).

   To reuse a project for other platforms or products, package the project as a VxWorks layer. A layer is also a good delivery mechanism for sharing components in a large project, or as a mechanism for contractors to deliver middleware or drivers for a project.

   For more information on layers, see the VxWorks 7 Layers and Package Management Guide.

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7. APPLICATION DEVELOPMENT WITH WORKBENCH 4

Wind River provides video tutorials, text-based tutorials, and sample project template programs to get started with application development using Workbench 4.

Workbench 4 User Interface

To get help on the Workbench graphical user interface from Workbench 4, navigate to **Help > Help Contents**, expand **Wind River Documentation** and then expand **Workbench 4** to access the *Workbench 4 User Interface Reference*.

The user interface reference describes text fields, icons, and other user interface elements of many Workbench 4 views and dialogs.

Workbench 4 Project Tutorials

The VxWorks 7 getting started videos and text-based tutorials provide guidance on basic Workbench 4 operations including the creation of VxWorks source build (VSB)projects, VxWorks image projects (VIPs), kernel applications, and real-time process (RTP) applications.

To access the tutorials, go to Getting Started and select **Tutorials/Videos**.

Workbench 4 User Tutorials

Workbench 4 user tutorials guide you through basic operations such as searching and replacing text, importing and exporting from Workbench Projects, and connecting to a virtual machine.

To access the user tutorials, go to Workbench 4 documentation and select **Getting Started**.

Workbench 4 Sample Project Template Programs

Workbench 4 provides sample project template programs to get started on project development. The sample templates include the following project types:

- Native
- Downloadable kernel module (DKM) application
- RTP application

To access the sample template projects from Workbench 4, navigate to **File > New > Example**.

The DKM application project files are at:

installDir **/vxworks-7/samples/dkm**

The RTP application project files are at:

installDir **/vxworks-7/samples/rtp**

Workbench 4 XML Sample Project

Workbench 4 provides a sample **XMLExamples** project with a collection of examples that illustrate how to use some of the editing features in the XML editor to edit and validate XML files.

To obtain an **XMLExamples** project from Workbench 4, navigate to **File > New > Example** and create the **XMLExamples** project.

A project folder named **XMLExamples** is created in the workspace directory.

Application Development

For information on creating, configuring, and building kernel application and RTP application projects, see the VxWorks 7 Kernel Application and RTP Application Projects Guide.

For information on developing kernel and RTP applications, see the *VxWorks 7 Application Programmer's Guide:RTP Applications* and *VxWorks 7 Programmer's Guide: About Real-Time Processes*.

8. PROJECT CREATION WITH THE WRTOOL UTILITY

VxWorks 7 provides the wrtool utility to create, build, and manage VxWorks projects from the command-line interface (CLI).

The wrtool utility can be used from a standard shell, or within Workbench 4 from a terminal window. When the wrtool utility is invoked, the workspace directory must be defined. You can either use the -data command option or the **WIND_WRTOOL_WORKSPACE** environment variable to specify the workspace directory.

The wrtool utility is located in *installDir* **/workbench-4**. For information on setting environment variables for the VxWorks development tools, see Environment Variables for the VxWorks Development Tools on page 14.

> Note: Although the wrtool utility and Workbench 4 can be used interchangeably to create, build, and manage VxWorks projects, the wrtool utility and Workbench 4 cannot use the same workspace at the same time.

Launching the wrtool Utility

To launch the wrtool utility, open a VxWorks development shell from the command-line interface or from Workbench 4 and use the wrtool command.

To view the startup command options for the wrtool utility, type wrtool /?. For example:

```
C:\WindRiver\workbench-4>wrtool
                    /?
                WrTool, version 4.1.0 Usage: wrtool [-debug] [-data
                    directory] [-port port] [-configuration directory] [file | cmd args...]
                    ...
```

To launch the wrtool utility with the -data command option, use the wrtool -data *directory* syntax. For example:

```
C:/WindRiver/workbench-4>
                    wrtool -data
                    C:\WindRiver\workbench-4\workspace
                Active workspace is
                    'C:\WindRiver\workbench-4\workspace'. WrTool, version 1.0 Copyright (c) 2013 -
                    2014 Wind River Systems, Inc.
                workbench-4> cd workspace
                workspace>
```

To launch the wrtool utility based on the worspace directory set with the **WIND_WRTOOL_WORKSPACE** environment variable, type wrtool. The active workspace is set based on the value of the **WIND_WRTOOL_WORKSPACE** environment variable. For example:

```
C:\WindRiver\workbench-4>wrtool
                Active workspace is
                    'C:\WindRiver\workbench-4\workspace'. WrTool, version 4.1.0 Copyright (c) 201
3 -
                    2014 Wind River Systems, Inc.
                workbench-4> cd workspace
                workspace>
```

Getting Help for wrtool

To view the wrtool command groups and general help commands, type help.

To get help for all wrtool commands, type help all.

To get help for the active wrtool session, type help builtin.

To get help with project commands, type help prj.

To get help for a specific command, type help followed by the command and command options. For example:

```
workspace>
                    help prj
                workspace> help prj vsb
                workspace> help prj vsb create
```

Command-Line Tutorials

The command-line tutorials provide guidance on basic operations including the creation of VxWorks source build (VSB) projects, VxWorks image projects (VIPs), kernel applications, and real-time process (RTP) applications using the wrtool utility.

To access the tutorials, go to Getting Started and select **Tutorials/Videos**.

Creating VSB and VIP Projects with the wrtool Utility

For information on configuring and building VxWorks source build (VSB) projects and VxWorks image projects (VIPs) using the wrtool utility, see the VxWorks 7 Configuration and Build Guide.

Using wrtool Commands in Scripts

When using wrtool commands in scripts, the workspace must be specified with the **WIND_WRTOOL_WORKSPACE** environment variable. Furthermore, to prevent the untimely shutdown of the wrtool session, set the **WIND_WRTOOL_KEEP_ALIVE_TIMEOUT** environment variable to keep a running instance of the wrtool utility throughout the execution of the script. Upon script completion, run the exit command to terminate the wrtool session.

9. ENVIRONMENT VARIABLES FOR THE VXWORKS DEVELOPMENT TOOLS

VxWorks environment variables are placeholders that specify the location of software resources for the VxWorks development tools.

The basic environment variables for the VxWorks development tools are as follows:

Environment Variable	Value Syntax
WIND_HOME	*installDir*
WIND_BASE	*installDir* **\vxworks-7**
WIND_WRTOOL_WORKSPACE	*installDir* **\workbench-4\workspace**
WRSD_LICENSE_FILE	*installDir* **\license***licenseFilename*